

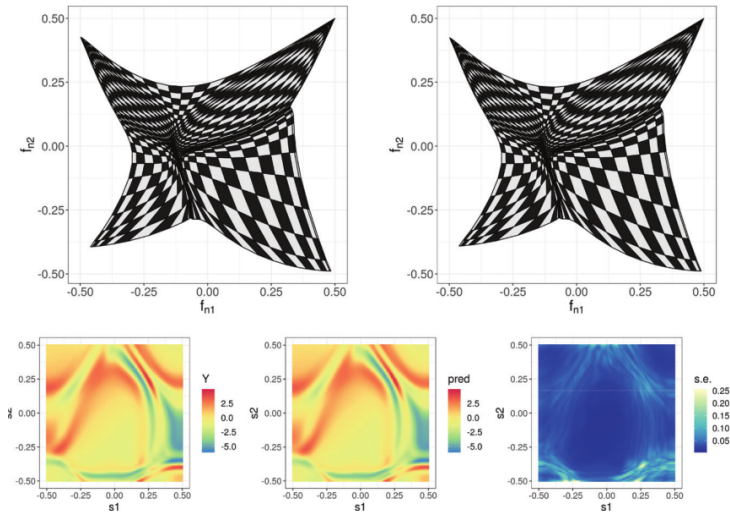
Roundtable Discussion II: Modeling Spatial Data and Extremes with Machine Learning

S. Engelke and A. Zammit-Mangion

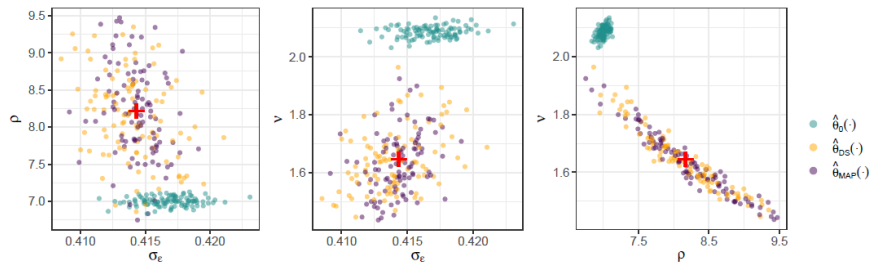
Objectives

- **Objective 1:** Discuss ways in which machine learning has been, and can, be used for the analysis of spatial/spatio-temporal data and extremes. What are the challenges?
- **Objective 2:** Discuss the limitations of ML - what are the ingredients of our statistical modelling approach that can be replaced and that cannot be replaced using ML?

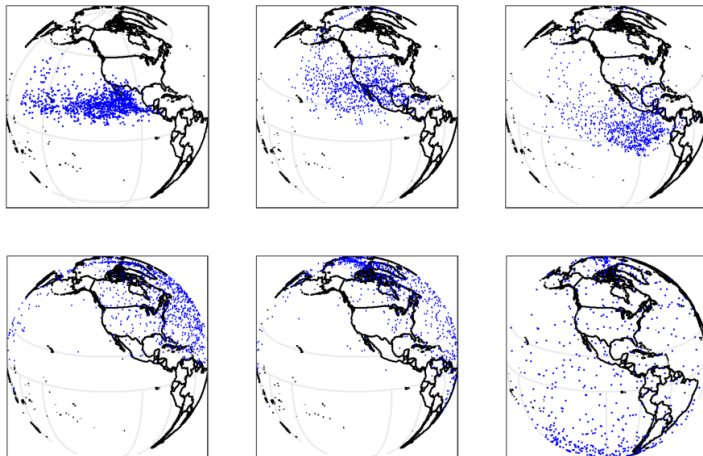
ML in Spatial Statistics: Process modelling



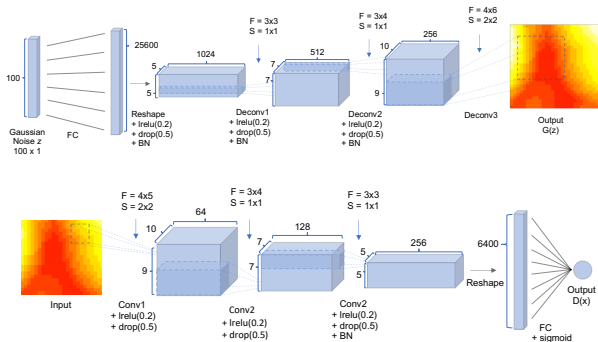
ML in Spatial Statistics: Neural Estimation



ML in Spatial Statistics: Intensity-function estimation

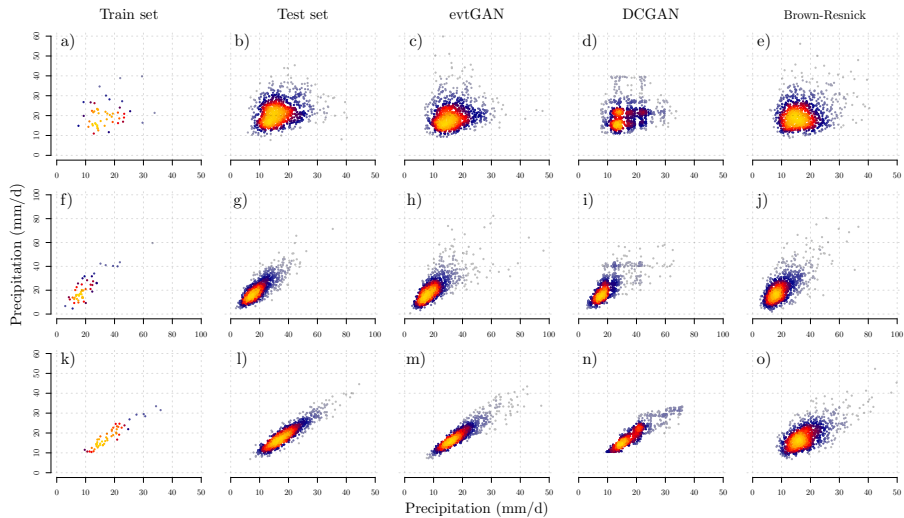


ML in Spatial Extremes: GANs for extreme event generation

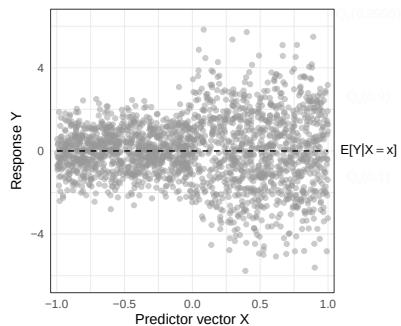


Boulaguiem, Zscheischler, Vignotto, van der Wiel & Engelke. *Modeling and simulating spatial extremes by combining extreme value theory with generative adversarial networks*. Environmental Data Science 1, 2022.

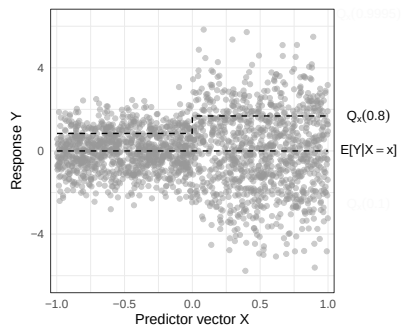
ML in Spatial Extremes: precipitation samples



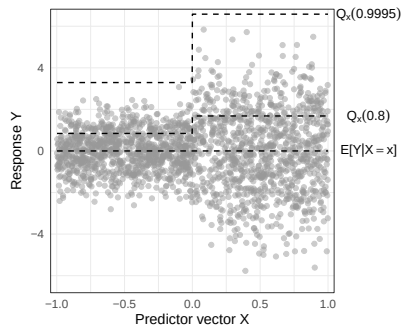
ML in Extremes: extreme quantile prediction



ML in Extremes: extreme quantile prediction



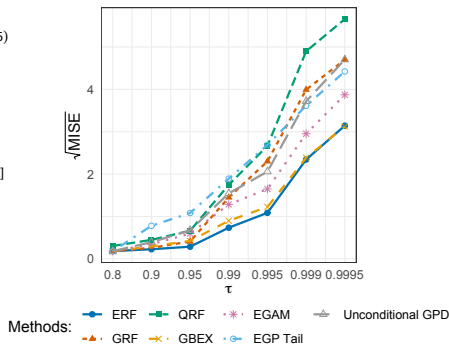
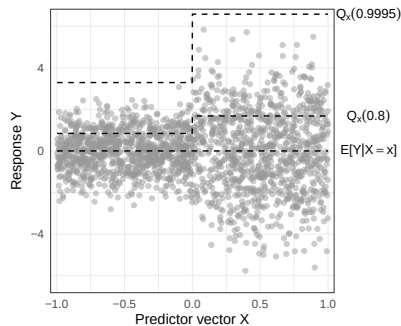
ML in Extremes: extreme quantile prediction



- GPD/GEV regression of conditional quantile at level τ close to 1:

$$Q_X(\tau) = F_Y^{-1}(\tau | X = x)$$

ML in Extremes: extreme quantile prediction



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Gnecco, Terefe & Engelke. *Extremal random forest*. (arxiv) 2022.

Objective 1 (1/2)

- What are some of the areas where ML can make a key difference when tackling spatial problems (extremes or not extremes)?
 - Flexible modeling of non-stationarity via deformations?
 - Flexible prediction methods on extreme properties (quantiles, etc.)?
 - What else?
- What are main challenges of ML in extremes?
 - Data scarcity?
 - Different loss functions?
 - Validation of models?

Objective 1 (2/2)

- How can ML be used
 - to help in the graphical structure learning of (sparse) multivariate / spatial extremes?
 - to aid detection and attribution?
 - to answer causal questions related to extremes?
- How can the following ML techniques be used to help us do better spatial (extremes) analysis:
 - Deep Learning models
 - Random Forests
 - Reinforcement learning
 - Other (as yet unexplored?)

Objective 2

- Where/what is the interface between ML (deep learning models) and traditional spatial statistics? That is, what are the things we do in spatial statistics that cannot be replaced by ML and what can ML do that spatial statistics can not? (Is uncertainty quantification one of these things?)
- How do we keep deep learning models & methods interpretable / explainable, and is this important to do in every situation?